Capturing the Value created by The Innovation Game's Algorithms

Introduction

With the announcement of a world beating algorithmic method for vehicle routing that shows very significant gains over the previous state of the art, The Innovation Game has proved itself as a system capable of effectively incentivising the creation of state of the art algorithms without compromising openness. This system now has value in its own right as one which can coordinate resources to produce valuable innovation.

A feature of the Innovation Game's design is that it is able to secure intellectual property rights for the innovation that the system generates and utilise those rights to simultaneously permit open innovation and monetisation. That The Innovation Game supports open innovation is essential to imbue it with the properties that (i) engage community enforcement of the intellectual property rights; and (ii) fosters open innovation; and (iii) the capture of value from that innovation to fairly and efficiently route it back to the value creators.

Followers of the Innovation Game have always identified The Innovation Game as being something different from typical tokenised enterprises because it is able to make the plausible claim that its output can generate sustainable revenue.

For The Innovation Game to succeed as envisioned there are 6 elements of the system that must be delivered.

1. A system that is secure and robust that can determine the **relative** performance of a group of algorithms.

2. A system that incentivises the improvement of the **absolute** performance of algorithms.

3. A system that captures intellectual property rights for the innovation that improves the absolute performance of algorithms and is capable of enforcing those rights.

4. A system that can simultaneously support value capture from commercial users of the algorithms and open innovation.

5. Challenge Designer Community: A system that incentivises the natural development of an ecosystem of solutions services providers that will drive the development of TIG challenges (perhaps in conjunction with university researchers) that are aligned with inducing algorithms which have real world utility and therefore value.

6. Innovator Community: A system that incentivises the natural development of an ecosystem of solutions services providers that will develop algorithms which solve TIG challenges (perhaps in conjunction with university researchers), but which will importantly also implement those algorithms in solutions customised for specific

customer requirements, effectively pulling through the output of The Innovation Game and making it useful for customers.

With The Innovation Game system having proven that it can generate meaningful innovation it is now appropriate to communicate more explicitly about how The Innovation Game will capture the value of its output.

Achieving Success for The Innovation Game through Licensing

Step 1: Identify your target customer (licensee)

The output of the Innovation Game is algorithm implementations. Some algorithm implementations are useful to end users with only limited customisation whilst others require higher levels of customisation because of the circumstance specific nature of their use. In other words, some algorithms output by the Innovation Game are more "*finished*" than others.

For algorithms which are useful with only limited customisation, end users may be an appropriate licensing target, but for those algorithms which require higher levels of customisation an intermediary that can do the customisation is the appropriate target.

Step 2: Understand the specification of an asset the licensee wants

The Innovation Game challenges are designed to deliver as much alignment between the induced solutions and end user utility as possible, but without requiring a different challenge for each circumstance specific use case. Finding the right level of abstraction for the challenge design is one of the key decisions that The Innovation Game has to make.

Step 3: Create the asset

There are three elements to technology asset creation: (1) the development of a technology; (2) the securing intellectual property rights on that technology; (3) the plausibility of enforcement of those intellectual property rights.

Technology Development
The Innovation Game is performing well with respect to inducing the creation of aligned algorithms, for example setting new standards for SOTA and BKS for Vehicle Routing. New challenges will be added to induce a broader range of algorithm advances over time.

Securing Intellectual Property Rights
To secure control over its algorithms, The Innovation Game must secure associated intellectual property rights.

Algorithms can be *effectively* protected by intellectual property rights.

> **Copyright**
> It is true that copyright (ordinarily the "go-to" intellectual property right to protect software) may provide only relatively weak protection against infringement for algorithmic methods because copyright protects the particular expression of an idea and not the underlying ideas expressed (the so called "idea/ expression dichotomy") and so is limited only to the specific implementation (i.e. the literal code itself) of an algorithmic method.

Because TIG publishes source code, we expect that it will become increasingly easy with the rapidly advancing capability of coding agents, to “extract” the ideas embodied in the published code and reimplement them in new code; effectively avoiding copyright protection. This makes patent protection for algorithms of primary importance if you want to protect them from being copied and used otherwise than in accordance with the permissions granted under The Innovation Game licences.

Patents
It is a common misconception that algorithms cannot be patented because, in most significant economic jurisdictions, algorithms *per se* are not patentable subject matter; the objection being that algorithms represent abstract ideas that should not be monopolised. However, there are ways around this exclusion by linking the algorithm to a technical effect or effects caused by execution of the algorithm. The filing of patents on the technical effects of algorithmic methods is, in fact, commonplace.

The procedural design of The Innovation Game allows it to secure inventions and patents.

Enforceability of the Intellectual Property Rights
Intellectual property rights are of little value if they cannot be enforced to prevent free riding.

Because a central pillar of The Innovation Game is open innovation, The Innovation Game has a powerful and, in principle, well established enforcement resource available to it that is uniquely available to open innovation ecosystems; “community enforcement”.

Because The Innovation Game must publish its algorithms to support open innovation, Community enforcement is key to the feasibility of preventing freeriding with respect to the output of The Innovation Game. If community enforcement is established, The Innovation Game will not need to rely on expensive legal enforcement through the courts.

For any form of enforcement to be effective, it must provide leverage against infringers. Enforcement leverage in the context of community enforcement primarily comes from the credible threat of **reputational damage** that would be suffered by an infringer, and the consequences of that reputational damage;

Consequences of reputational damage include:

- **Loss of technical cooperation**: Most would-be infringers will have some reliance on the technical cooperation of resources in the community (including reliance on access to academic researchers and to tacit knowledge of experts more generally).

- **Loss of innovation resources**: The commercial community whose objective it is to see TIG thrive in order to maintain access to open algorithms that might otherwise be closed and monopolised (particularly those that improve AI models) will withdraw support from and may be actively hostile towards free riders.

- **Consumer ostracization**: The crypto community is very adept at amplifying messaging across social media and could be effective in bringing reputational damage to a consumer facing free rider.

- **Access to public funding/policy influence**: Governments may see TIG as an alternative or supplementary funding source to their own funding of science and technology and wish to promote and sustain TIG. Governments may also see TIG as a component of the means by which the monopolisation of algorithms is avoided. Governments may withdraw funding from free riders or be unsympathetic to their policy concerns.

In addition to the reputational risk, the other significant risk for a free-rider is that they may also be exposed to the asymmetric risk of an injunction and/or, where they have used code obtained under the Open Data Licence, specific performance of the data disclosure and source code share alike obligations in that licence.

Appendix A explains in more detail how community enforcement works, and why it can plausibly operate as an enforcement mechanism for The Innovation Game's intellectual property.

Step 4: License the asset to the licensee whilst still maintaining the openness of the algorithm

Appendix B sets out the various licence types operating in The Innovation Game. They essentially fall into three categories; (i) Game Licences; (ii) Open Licence; and (iii) Monetisation Licences

Simultaneous value capture from commercial users and open innovation
The Innovation Game has as its objectives (i) the fostering of open innovation; and (ii) the capture of value from that innovation to fairly and efficiently route it back to the value creators.

Preserving Openness
To enable open collaboration, The Innovation Game always makes implementations of algorithms submitted to The TIG Foundation available under The TIG Open Data License (a licence that meets objective standards of openness):

> ***The TIG Open Data Licence:*** A licence inspired by the CERN open hardware licence and which includes a share alike obligation (meaning that any derivative of the licensed code must be relicensed under the Open Data License only).

The TIG Open Data License requires that if data (including implementations thereof) generated by execution of the subject algorithm implementation ("**Output Data**") on certain input data is made available to third parties as data or in the form of a product derived from the data, then the input data processed by execution of the implementation to generate the Output Data together with any additional information to generate a product derived therefrom must also be made **generally available** to the extent necessary to allow a reproduction of the Output Data or product as the case may be (such input data is referred to as "**Relevant Data**").

The share alike obligation will require that; **(i)** where the subject algorithm implementation alone is distributed, the source code for the implementation must be made available under the TIG Open Data License **to the recipient** of the algorithm implementation; and **(ii)** where Output Data is distributed, the source code for the implementation and the Relevant Data must be made **generally available** for distribution under the TIG Open Data License.

The design objective of the Open Data Licence is that the licence terms are completely compatible with the reasonable requirements of the academic community (allowing them to continue to use TIG algorithms as the basis for further research and development without the terms of the license restricting them) WHILST being incompatible with most commercial interests. It is expected that for most commercial enterprises the data disclosure requirements in that license will provide an incentive (i.e. the avoidance of the data disclosure requirements) for those commercial enterprises to alternatively take a commercial license for a fee which will then be channelled back to the original innovators.

Capturing Value

For algorithms that are "finished" the appropriate commercial licensee will be an end user, and they will access the algorithm under the ***TIG Commercial Licence***.

> ***The TIG Commercial Licence:*** A licence which will allow closed use of algorithms and Relevant Data in return for the payment of a fee (non-discriminatory, means tested and rate card based) which will be used to underpin value in The Innovation Game incentives. In contrast with the TIG Open Data License, *The TIG Commercial License* will provide greater freedom with respect to downstream licensing and will not impose an obligation to make data available.

For algorithms where there is "customisation gap" we are currently working with intermediaries (so called "integrators") who will provide a customisation service for end users. The Innovation Game incentives already operate to encourage the emergence of an ecosystem of such integrators and additionally they will be able to generate revenue by charging for their customisation services. For algorithms that require customisation the appropriate licensee will be an integrator, and they will access the algorithm under the ***TIG Integrator Licence***. With the exception of distribution to The Innovation Game under the TIG Inbound Game Licence, the integrator licence limits the distribution of TIG algorithms and their derivatives solely to entities which have entered into a TIG Commercial Licence.

> ***The TIG Integrator Licence:*** A licence which will allow use of algorithms and Relevant Data by the integrator for the purpose of customising the algorithms for end users together with a limited right to distribute the algorithm to such end users provided that they entered into a TIG Commercial Licence.

Pricing of Commercial Licenses

The leverage resulting from community enforcement of intellectual property rights will always be evaluated by would be infringers against the cost of taking a licence form TIG.

There is no established commercial market for algorithms because algorithm development has to date been largely publicly funded and open. As a result, there is no precedent for the commercial licensing of unpackaged algorithms and therefore no pricing precedent.

Even if there was pricing precedent for unpackaged algorithms, the massive value of algorithms in the most fundamental innovation of all, AI, means that any data from a previously existing market would have been unlikely to have provided a useful precedent.

Price discovery can only happen in a real market and will be dynamic depending on:

- The quality of assets available for license; including how “finished” the algorithms are and their relevance and commercial value
- The quantum of assets being licensed
- The efficacy of The Innovation Game as an innovation system and commercial reliance thereon
- The enforceability of the relevant intellectual property rights

TIG proposes, ultimately, to make all of its algorithms available as a single bundle under the commercial license subject to consideration determined by a non-discriminatory, means tested rate card which will be set by The Innovation Game but which can be varied from time to time. Ultimately it is intended that the rate setting will be decentralised to token holders. If TIG is successful as an “innovation system” the price should increase over time to reflect the value added as innovation proceeds.

Conclusion

The Innovation Game has proven that it is a system capable of incentivising innovation that advances the state of the art provided that the value of the token incentive can be sustained at the current level or above. In fact, the higher the token value the more innovation will be created by the “innovation flywheel”.

We are in the process of establishing a low friction system for securing Intellectual Property Rights from Universities and other research communities.

Whilst we have succeeded in aligning the TIG challenges to provide a good starting point for commercial use cases; we must develop an ecosystem of integrators that can take the algorithmic methods and implementations generated by TIG and customise them for individual customers to close the “customisation gap”; effectively the “last mile” in completing the route to market for TIG algorithms.

We need to foster a community that will embrace “community enforcement”; there is a close analogue between The Innovation Game and open source software, where it has been observed to operate.

If the above things can be fulfilled, then the licensing of the resulting IP presents no novel difficulties: we are back in the orthodox world of technology licensing at that point.

Appendix A
Community Enforcement

Naturally, to rely on community enforcement you need a community to provide the enforcement.

For a community to form and develop an enforcement motive it must be seeded and sustained by**; (i)** a cause, **(ii)** a means to act; and **(iii)** perceived legitimacy in acting.

TIG's "Cause"

- Open innovation: the preservation of openness and collaboration around ai development. keeping frontier algorithmic methods open.
- Fair reward for access: market determined reward for algorithmic innovation provided that it is open.

AI-assisted algorithm development has now emerged as feasible and is becoming increasingly effective (state-of-the-art results are now developed with AI), leading to the prospect that an AI can be leveraged to self-improve. Such recursive improvement in the hands of a single entity could potentially lead to monopolisation of algorithm development by that entity.

Currently, competitive AI-assisted algorithm development remains reliant upon the intervention of human insight and intuition. Much of the required human insight and intuition is tacit (never written down but nevertheless relied upon for problem solving) human knowledge yet to have been significantly captured in any recorded dataset available to AI, but which can be divined by AI-assistants from their interactions with experts. Capture of those interactions with experts is accordingly extremely valuable information and the ability of an AI assistant to incentivise such interactions and capture the information that flows from them, can potentially lead to compounding network effects (a "data flywheel") akin to those that gave Google a 25 year monopoly on Search. Such a data flywheel would be likely to lead to monopolisation of AI assisted algorithm development by the controller of the AI assistant around which the network forms.

AI-assisted algorithm development necessarily requires significant investment because the development process is very compute intensive. To finance the process, there is a natural incentive for the controller of the AI assistant to keep the algorithms output by the assistant **closed** because in the absence of effective intellectual property protection there would be no clear route to monetisation if they were published openly.

The "Means to Act"

The means to act is provided by **The Innovation Game (TIG):** A protocol that provides monetary rewards for developing more performant algorithms to solve important problems in science, engineering and AI, whilst allowing for those algorithms to remain open.

Importantly, TIG rewards are allocated by a market mechanism. A mechanism for fairly allocating rewards was noted by Eric Raymond in *The Magic Cauldron* to be the missing piece for open source principles to be more generally applicable.

“Perceived Legitimacy”

Members of a community need to hold a perception that, in pursuit of the community’s cause, they will be acting legitimately. Legitimacy for a community is based on Legal Legitimacy and Social/Normative (non-legal) legitimacy.

Legal Legitimacy

- Intellectual Property or other legal rights that are plausibly valid and plausibly infringed.

Social/Normative Legitimacy

- **TIG enables fair allocation of reward for effort**: The TIG market for algorithms provides "impersonal" allocation, which reduces the risk of community fracture (intentional allocation tends to result in perceptions of injustice sooner or later, as per Hayek).

- **TIG is a Public Good**: Open Innovation provides access to all-comers and collaborative research is proven to be an efficient and effective methodology for fostering innovation.

- **TIG has a public interest mission**: The aversion of the threat of monopolised AI development and access.

- **Commercial self-interest of chasers to sustain competition**: Consensus that an open model is needed to combat dominance by one entity. Recall the precedent of Linux gaining wide corporate patronage once supporting it was identified as a defensive move to block a Microsoft monopoly. The same dynamic can operate with TIG as a defensive move against Google attaining a monopoly over algorithms.

- **Open Economic participation:** AI-assisted algorithm development can become a scaled industrial activity akin to Bitcoin mining, in which anyone can participate without permission.

Appendix B
TIG Licence Types

Game Licences
Licences to facilitate playing The Innovation Game

TIG Inbound Game Licence:
- Licence for contributions to TIG
- Applied by contributors to TIG Algorithm contibutions

TIG Outbound Game Licence:
- Licence for TIG algorithms to create derivatives for submission to TIG
- Applied by TIG to TIG algorithms in TIG GitHub monorepo

TIG Bench marker Licence
- Licence for TIG algorithms used by benchmarkers for benchmarking
- Applied by TIG to TIG algorithms in TIG GitHub monorepo

Open Licence
Licence to ensure that TIG algorithms stay open

TIG Open Data Licence (TODL)
- Applied by TIG to TIG algorithms in TIG GitHub monorepo

Monetisation Licences
Licenses to facilitate monetisation of TIG algorithms

TIG Commercial Licence
- Applied by TIG to TIG algorithms in TIG GitHub monorepo
- Licence to commercial end users

TIG Integrator Licence
- Applied by TIG to TIG algorithms in TIG GitHub monorepo
- Licence to integrators with rights to distribute to TIG Commercial Licensees